

RTRL Flight Controller: Results

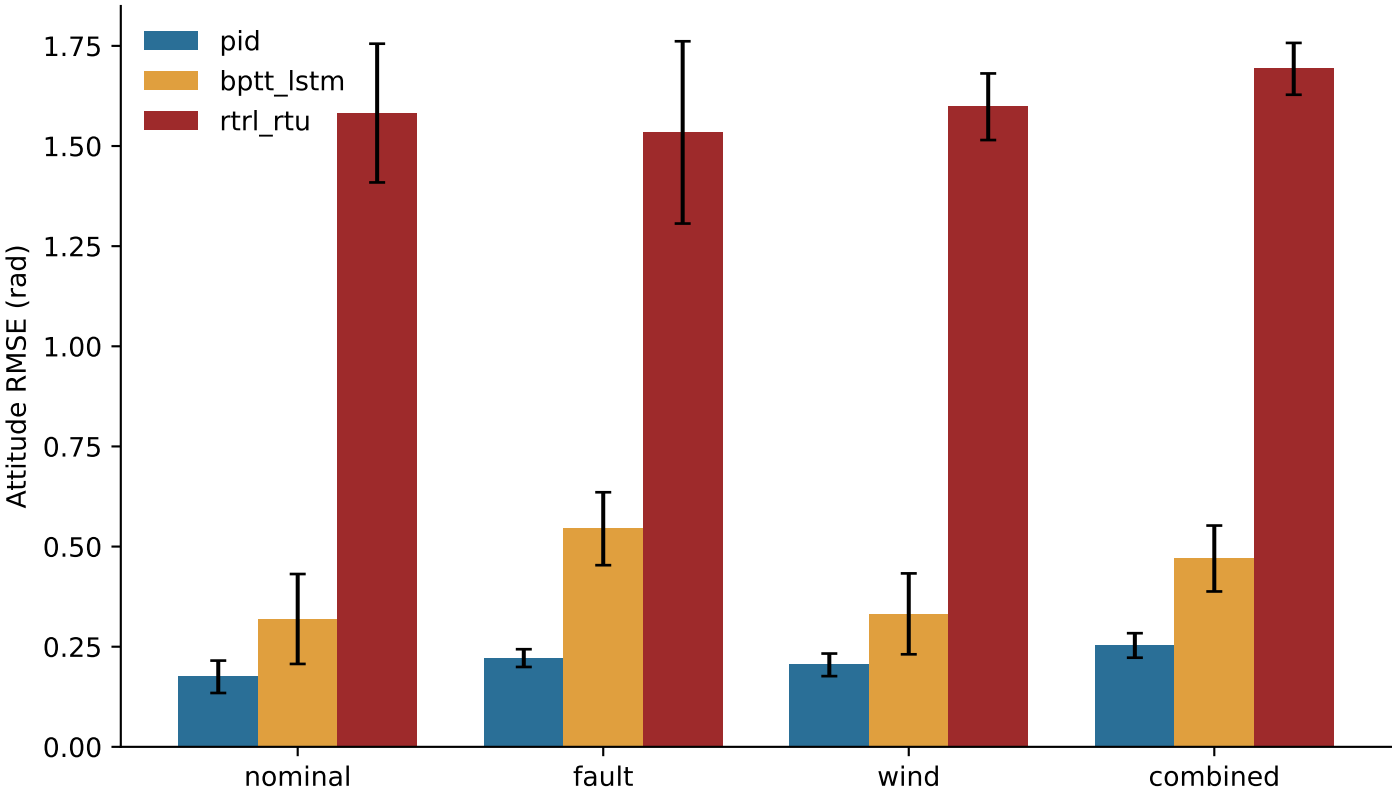
Cessna 172 in JSBSim, attitude hold. Controllers: PID, BPTT-LSTM (frozen), RTRL-RTU (online adaptation). Scenarios: nominal, actuator fault, Dryden wind, combined, plus an ablation (RTRL with online updates off). 3 seeds per cell, 39 episodes total. Source: data/results/sweep_results.parquet.

Headline

The project success criterion was NOT met. RTRL did not recover from actuator faults, and did not track attitude acceptably even in calm air. PID was best in every scenario, the frozen BPTT-LSTM second, RTRL-RTU far worse than both.

Scenario	Controller	Attitude RMSE (rad)	Mean return	Jerk
nominal	pid	0.175 ± 0.040	-10	0.0038
nominal	bptt_lstm	0.319 ± 0.112	-33	0.0017
nominal	rtrl_rtu	1.582 ± 0.173	-1357	0.0788
fault	pid	0.222 ± 0.022	-15	0.0044
fault	bptt_lstm	0.545 ± 0.091	-91	0.0017
fault	rtrl_rtu	1.534 ± 0.228	-1070	0.0684
wind	pid	0.205 ± 0.028	-14	0.0323
wind	bptt_lstm	0.332 ± 0.101	-36	0.0031
wind	rtrl_rtu	1.598 ± 0.083	-1073	0.0721
combined	pid	0.253 ± 0.031	-20	0.0317
combined	bptt_lstm	0.470 ± 0.082	-68	0.0027
combined	rtrl_rtu	1.693 ± 0.065	-1186	0.0688
ablation	rtrl_rtu	2.048 ± 0.262	-3720	0.1021

Attitude RMSE by scenario (error bars = std over 3 seeds)



Conclusion: what the results actually show

1. Ranking

PID < BPTT-LSTM << RTRL-RTU on attitude RMSE in all four scenarios. Nominal: PID 0.175, BPTT 0.319, RTRL 1.582 rad. RTRL is about 9x worse than PID, against a target of within ~15% of PID. An RMSE near 1.6 rad means the aircraft is not holding attitude at all.

2. The fault result (the core claim) is not supported

The hypothesis was that RTRL would recover from mid-flight actuator degradation where a frozen BPTT-LSTM would not. Fault RMSE: PID 0.222, BPTT-LSTM 0.545, RTRL 1.534, so RTRL is about 2.8x worse than the frozen LSTM. RTRL fault RMSE (1.53) is about the same as its nominal RMSE (1.58), so the fault is not what hurts it; it was already failing before the fault. No recovery time was measured for RTRL (N/A). The frozen LSTM does degrade under the fault (0.319 to 0.545), so the fault is a real disturbance, and PID handled it best (0.175 to 0.222).

3. Ablation: not evidence for online learning

With online updates off, RTRL RMSE is 2.05 ± 0.26 versus 1.53 ± 0.23 with them on (fault setup). Online-on looks better, but this rests on 3 seeds, both numbers are poor, and one ablation seed had a return of -8112, so variance is huge. It cannot be reported as "online adaptation helps recovery". At most it suggests the online update is not what breaks the controller.

4. Wind and combined

Same ordering. PID 0.205 / 0.253; BPTT 0.332 / 0.470; RTRL 1.598 / 1.693 (wind / combined). Wind adds little for any controller, so it did not discriminate between them.

5. Caveats

- Only 3 seeds per cell. The big ordering is robust; small differences are not meaningful.
- time_to_recover is mostly N/A. The two values present (BPTT 0.0 in fault, PID 73 in combined) are artefacts of the recovery definition and should not be interpreted.
- settling_time was not computed for any run.
- RTRL jerk (~0.07) is 20-40x higher than the other controllers, consistent with saturated or oscillating commands, not smooth adaptation.

6. Bottom line

In its current state the RTRL-RTU controller does not work as a flight controller: it fails nominal tracking before any fault or wind is applied. The fault-recovery claim is therefore untested rather than disproved, because the baseline competence needed to test it is missing. PID is the strongest controller and the frozen BPTT-LSTM is a credible learned baseline. A sensible next step is to find why the warm-started RTRL policy diverges from the imitation policy at deploy time (e.g. compare its first-step actions with the BPTT-LSTM) before drawing any conclusion about online adaptation itself.